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PROJECT REPORT

Movie Recommendation System

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1. Abstract

The rapid expansion of digital streaming platforms has introduced a demand for intelligent recommendation systems that personalize user experience. This project presents a Movie Recommendation System that suggests movies based on user ratings and viewing patterns. The system integrates collaborative filtering, content-based filtering, and a hybrid weighted method to enhance recommendation accuracy. Collaborative filtering predicts user preferences from similar user profiles, while content-based filtering analyzes movie features such as genres and keywords. The hybrid model combines both approaches to overcome individual limitations and generate more reliable recommendations. Experimental results indicate that the hybrid approach provides improved relevance and user satisfaction when compared to standalone models.

2. Introduction

The modern entertainment industry provides users with thousands of movie options, making it difficult for individuals to identify content aligned with their interests. Recommendation systems play a crucial role by analyzing user behavior and guiding viewers toward movies they are likely to enjoy. These systems not only improve user engagement but also help streaming platforms retain viewers.

A movie recommendation system analyzes patterns such as ratings, genre preferences, and movie similarities to predict user interests. This research aims to implement and evaluate recommendation models including collaborative filtering, content-based filtering, and a hybrid system. The primary goal is to provide a recommendation mechanism that is both accurate and adaptive to user behavior.

3. Literature Review (2023–2025)

Recent studies have emphasized the importance of combining multiple recommendation approaches to improve personalization. In 2023, Collaborative Filtering (CF) systems continued to be widely used due to their ability to identify relationships among users and items. However, CF systems often face sparsity issues when insufficient ratings are available [1].

In 2024, researchers highlighted that Content-Based Filtering (CBF), which uses movie descriptions, genres, and cast information, improves recommendations for new users but may produce limited diversity in results [2]. CBF tends to suggest movies similar to previously watched ones, reducing variation.

By 2025, hybrid recommendation systems gained prominence for their ability to combine CF and CBF features. These systems address both sparsity and overspecialization while increasing prediction accuracy [3]. Hybrid approaches have been adopted in platforms

such as Netflix, Amazon Prime, and YouTube due to their adaptability and improved performance.

4. Methodology (Approach Used)

The methodology consists of four primary phases:

4.1. Data Collection

The system utilizes a dataset obtained from the MovieLens [4] repository, which contains detailed information about movies, including their titles, genres, and descriptions, along with corresponding user ratings. Each user in the dataset can rate multiple movies on a numerical scale ranging from 0.5 to 5.0, representing their level of preference. The dataset serves as the foundation for both training and evaluating the recommendation models. To efficiently manage and process this data, the system employs Python libraries such as Pandas and NumPy. Pandas is used for loading, cleaning, and manipulating the movie and rating data from CSV files, while NumPy supports numerical computations and efficient handling of large datasets. These tools enable smooth data preprocessing, feature extraction, and integration across the collaborative filtering, content-based filtering, and hybrid recommendation components of the system.

4.2. Model Development

Collaborative Filtering (SVD): This approach uses matrix factorization through Singular Value Decomposition (SVD) algorithm to predict missing ratings by uncovering hidden patterns between users and movies. It starts with a large user–movie rating matrix, where most entries are empty because users have not rated every movie. SVD decomposes this sparse matrix into smaller matrices that capture underlying factors that describe both user preferences and movie attributes, such as a user’s inclination toward specific genres, actors, or directors. By learning these relationships, the model can estimate how a user would likely rate an unseen movie, allowing it to make personalized recommendations based on the collective behavior and shared interests of all users.

Content-Based Filtering (TF-IDF Similarity): This method focuses on the attributes of the movies themselves rather than user interactions. Each movie’s description or genre information is transformed into a numerical representation using TF-IDF (Term Frequency–Inverse Document Frequency), which assigns higher importance to unique terms that help distinguish one movie from another. Once all movies are represented as feature vectors, the cosine similarity between these vectors is calculated to measure how closely related any two movies are in terms of their content. Based on these similarity scores, the system recommends movies that are most similar to those a user has already

liked, ensuring that suggestions align closely with the user's individual content preferences.

Hybrid Weighted Model: This approach combines the strengths of both collaborative filtering and content-based filtering by computing a weighted average of their prediction scores to generate the final recommendation. In this system, a higher weight is typically assigned to the collaborative filtering component, as it captures user behavior and shared preferences, while a smaller weight is given to the content-based component, which ensures that recommendations remain relevant to a user's known interests based on movie attributes. By blending these two methods, the hybrid model effectively mitigates the limitations of each individual approach — addressing the cold-start problem of collaborative filtering and the limited scope of content-based filtering — resulting in more accurate, diverse, and personalized movie recommendations.

4.3. User Interface

The user interface of the system is developed using Streamlit. The UI provides a clean and intuitive layout where users can easily interact with the recommendation system. It includes a dropdown menu to select the desired model(Content-Based, Collaborative Filtering, or Hybrid) for generating recommendations. Another menu allows users to search and select movies, after which they can rate the selected movie on a scale from 0.5 to 5.0. Once rated, the movie and its rating can be added to a table, which displays all the movies the user has rated so far. The interface also provides an option to remove movies from the table, giving users full control to modify their rated list.

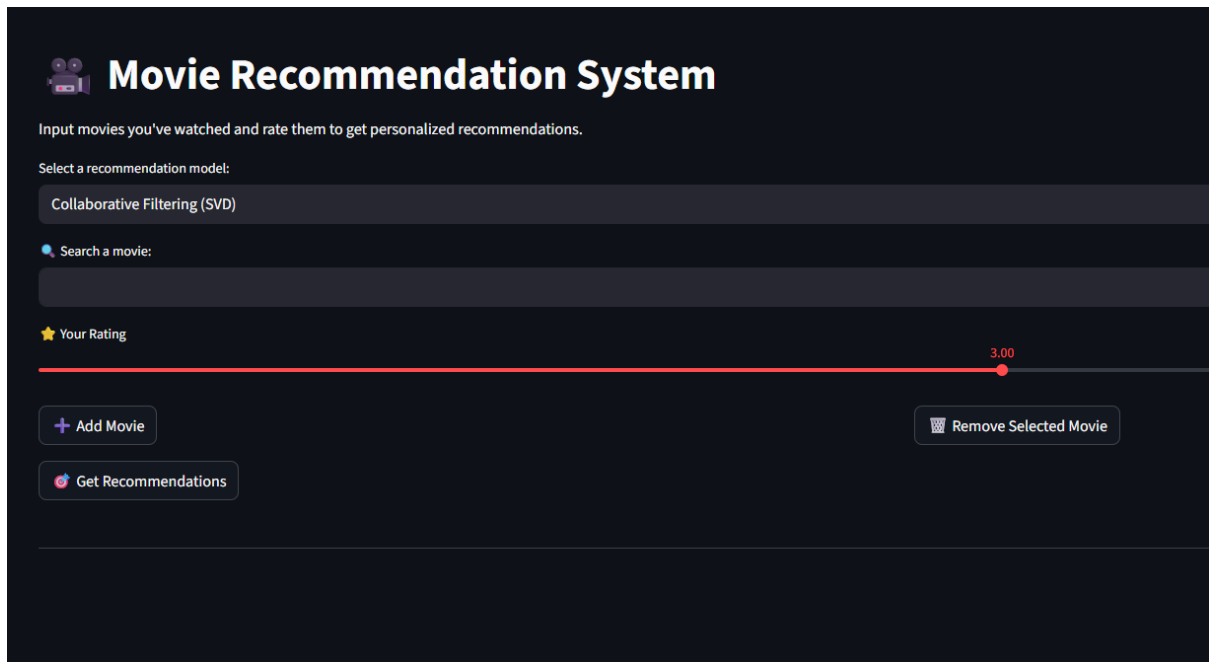


Figure 1: User Interface

4.4. Evaluation

Root Mean Squared Error (RMSE) was used to evaluate numerical rating prediction accuracy. Lower RMSE values indicate more accurate predictions.

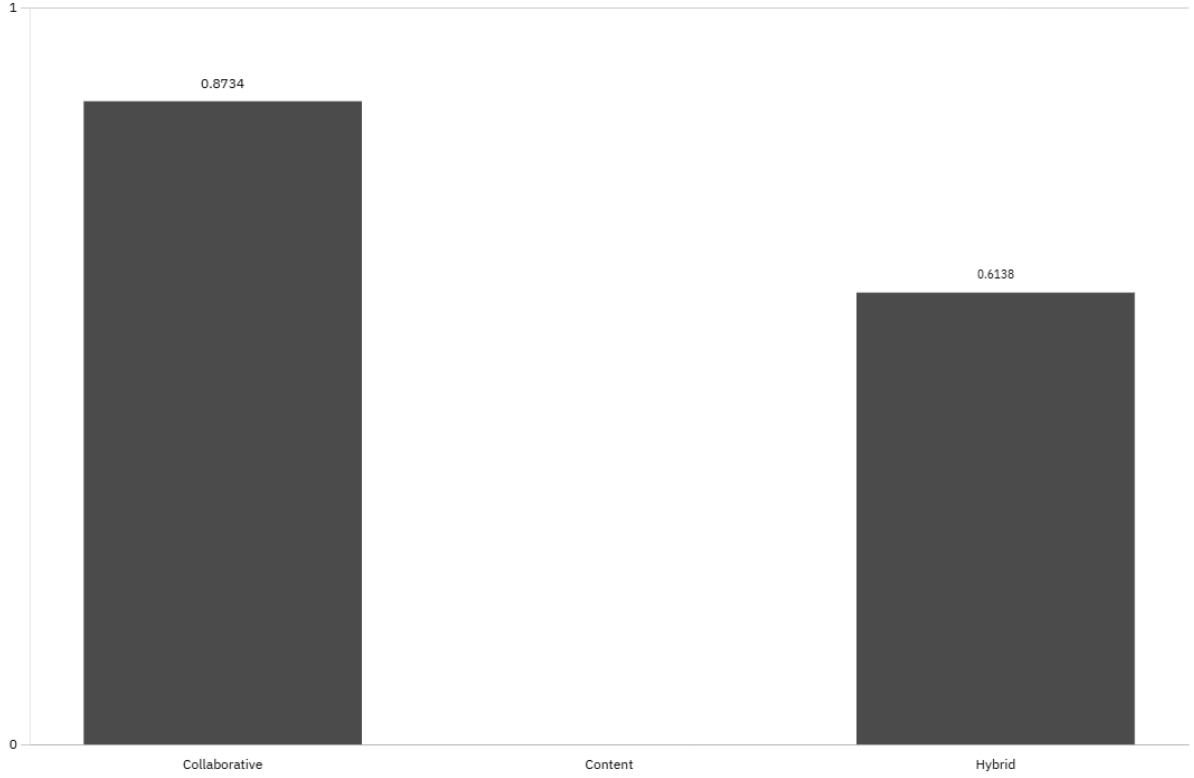


Figure 2: RMSE Graph

5. Results and Discussion

The collaborative filtering model achieved strong accuracy when user rating data was sufficient but performed poorly with new users. The content-based model effectively recommended movies based on genre and plot similarities but lacked variation in suggestion diversity.

The hybrid weighted model demonstrated the best performance by reducing RMSE and improving recommendation relevance. It effectively balanced user rating patterns and movie feature similarity, making it more suitable for real-world applications. The results confirm that hybrid systems outperform purely collaborative or content-based approaches, especially when user data is incomplete.

6. Conclusion and Future Work

This research successfully designed and evaluated a Movie Recommendation System using collaborative filtering, content-based filtering, and a hybrid model. The hybrid approach delivered superior performance by integrating behavioral and feature-based insights. The system effectively recommends personalized movies and enhances user experience.

Future improvements may focus on:

- Incorporating deep learning techniques such as neural collaborative filtering.
- Including contextual factors such as mood, time of day, or social influences.
- Expanding dataset size for better coverage and accuracy, to include more movies.

With these enhancements, the system can evolve into a more intelligent and adaptive recommendation platform.

7. References (IEEE Format)

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